

10.6 Assignment 10: Explainable artificial intelligence (XAI)

1. Why would you want transparency or explainability in an ML/AI system?

I think of an AI system like a friend giving me advice. If the friend refuses to explain why they're telling me to do something, I probably won't trust them. AI works the same way.

Advantages to users and the organization:

- It builds trust. It allows users to challenge mistakes.
- It gives people a sense of control. When they understand the logic, they are less fearful of it.
- It helps catch bias or unfair behavior and prevents reputational damage or legal trouble from discriminatory decisions.

2. What is a “fair” system?

A fair system treats people in a way that doesn't punish or favor them unfairly because of sensitive characteristics like race, gender, age, or disability. In simple terms fairness means similar people are treated similarly and no group is systematically disadvantaged.

For example, let's imagine an AI that decides who gets a home loan.

Unfair:

If two people with similar incomes and credit scores get different decisions because one belongs to a minority group.

Fair:

The AI makes decisions only based on relevant financial factors and removes any hidden correlations with sensitive features, like ZIP code indirectly reflecting race (a bias source mentioned in the document).

3. How are bias, fairness, accountability, and transparency related?

Bias

Bias is when the system's decisions are pushed in one direction because of bad or skewed data which usually happens if you train a model using incomplete data

Fairness

Fairness is the goal: reducing or removing that bias so all groups are treated equally.

Transparency

Transparency is how we see what the model is doing. If you can't see how the AI works, you can't detect or fix bias.

Accountability

Accountability is making sure someone is responsible for spotting problems, fixing them, and explaining decisions.

Here is a real life example

A school's entrance AI picks students.

- **Bias:** The data favored students from wealthier neighborhoods.
- **Transparency:** Helps reveal this.
- **Fairness:** Adjusts the model so all backgrounds get equal opportunity.
- **Accountability:** The school takes responsibility for mistakes and corrects them.